Name: Vanshika Wadhwani
Class : D20A
Roll no: 35

Experiment: 9

Aim:

To create, compile, deploy, and test an ERC-20 token using Solidity in Remix IDE and deploy the token on the Sepolia Ethereum test network using MetaMask.

Theory:

An ERC-20 token is a fungible token standard on the Ethereum blockchain. Fungible means that every unit of the token is identical and has the same value as every other unit of the same token. ERC-20 defines a common interface that allows wallets, exchanges, and smart contracts to interact with tokens in a standard way.

The main ERC-20 functions include totalSupply(), balanceOf(), transfer(), approve(), allowance(), and transferFrom(). Important events include Transfer and Approval.

In this experiment, an ERC-20 token named EXP09 Token, with the symbol EXP, was created using OpenZeppelin's ERC-20 implementation. An initial supply of 1000 EXP was minted to the deploying account. The contract was compiled and deployed on the Sepolia testnet through MetaMask. The deployed contract functions were then executed to verify the token's properties, total supply, account balances, and token transfers.

Program:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

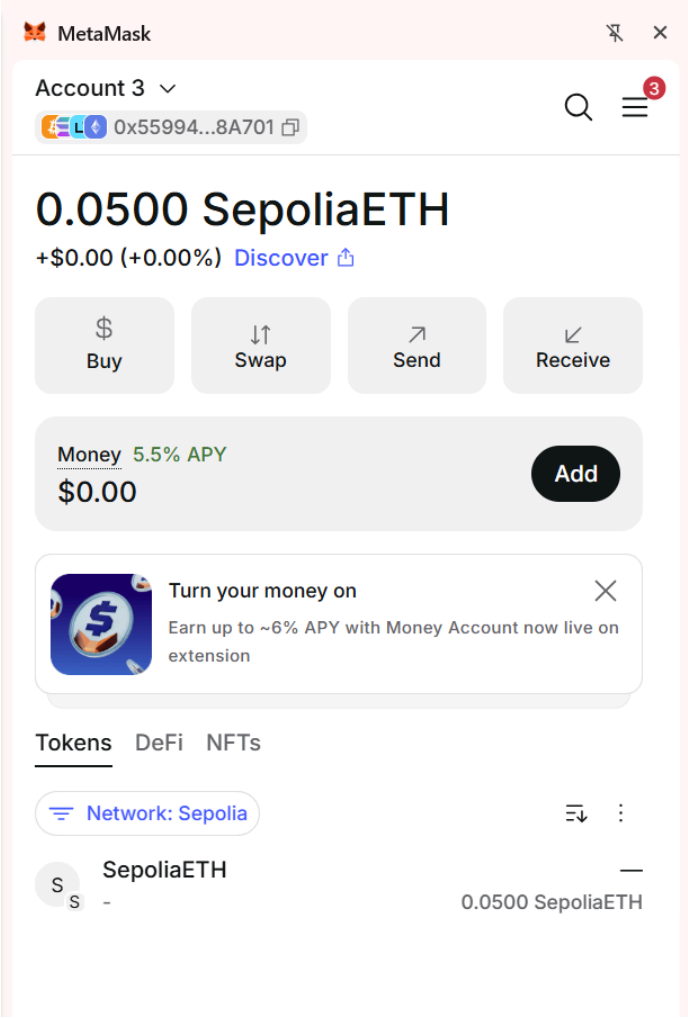
import {ERC20} from
   "@openzeppelin/contracts/token/ERC20/ERC20.sol";

contract Exp09Token is ERC20 {
   constructor(uint256 initialSupply)
      ERC20("EXP09 Token", "EXP")
   {
      _mint(
         msg.sender,
         initialSupply * 10 ** decimals()
```

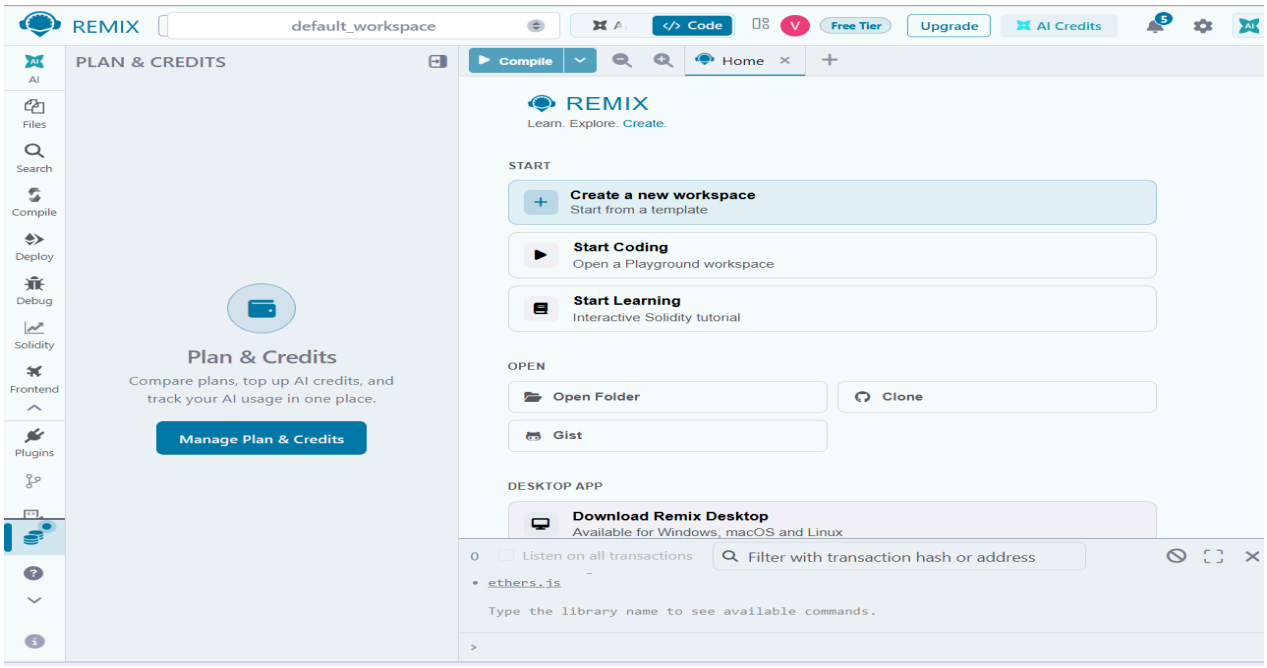
```
        );
    }
}
```

Output / Results:

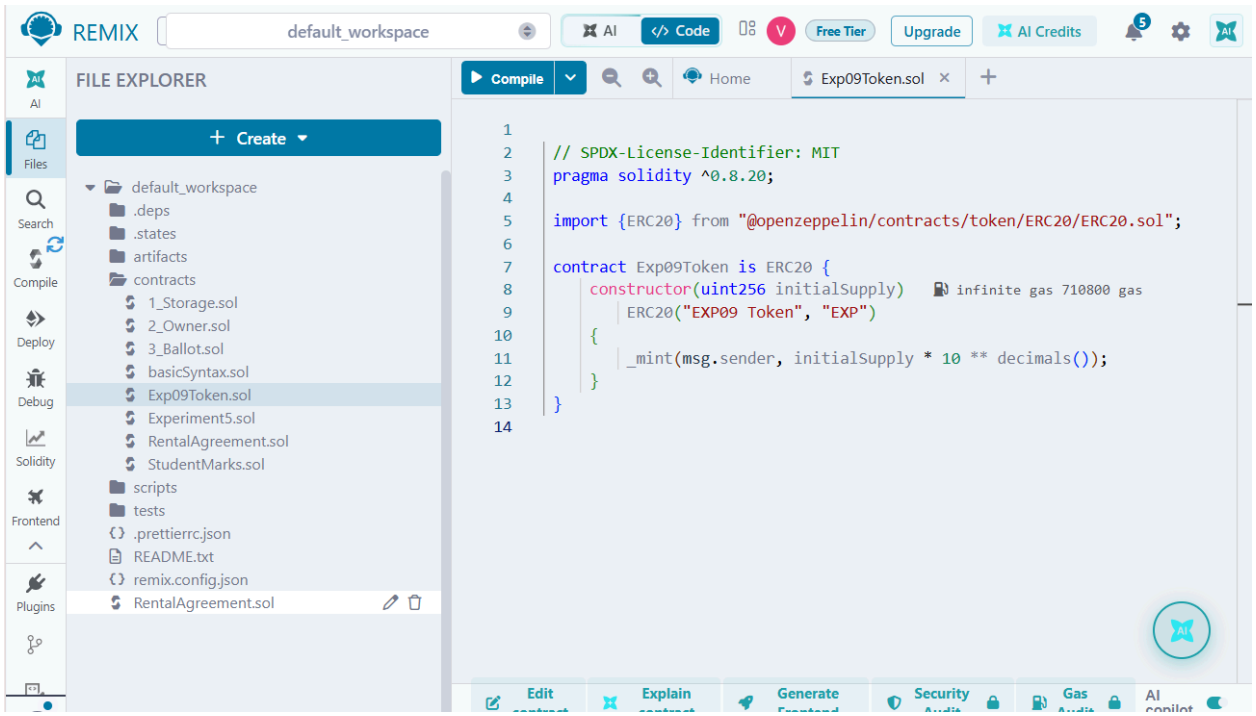
Metamask Wallet and Sepolia test ETH



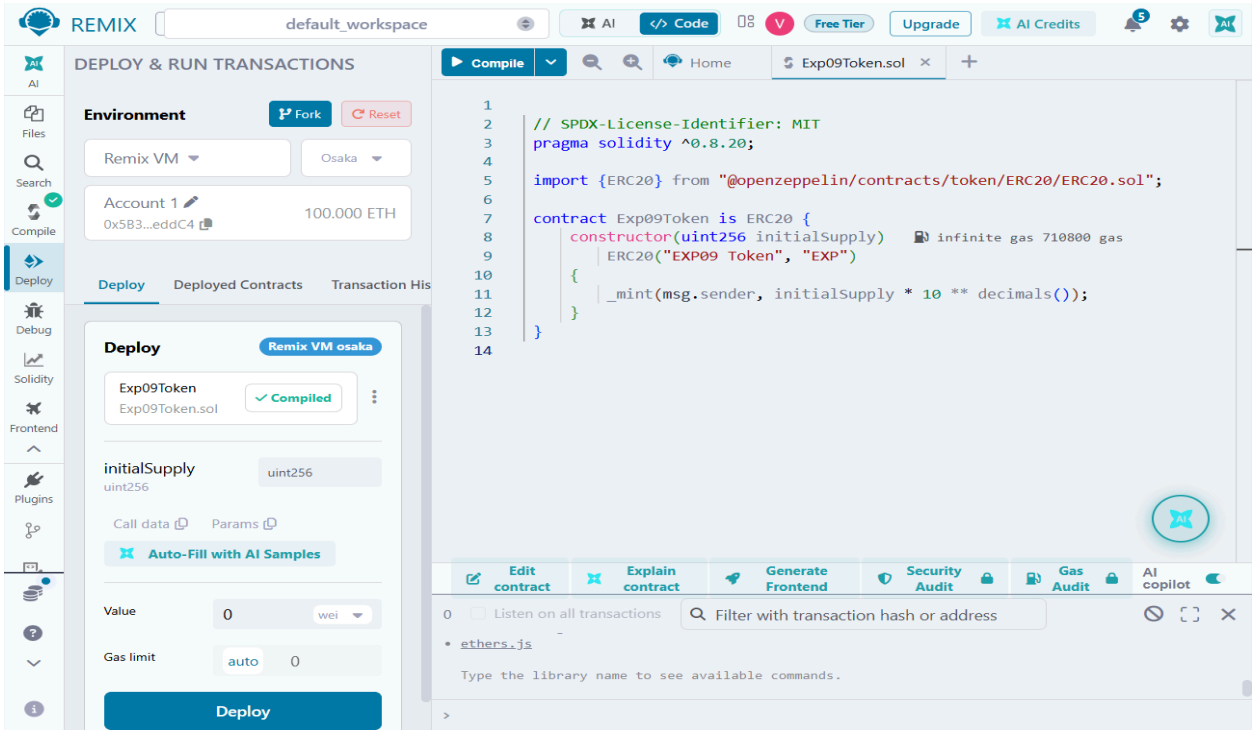
RemixIDE:



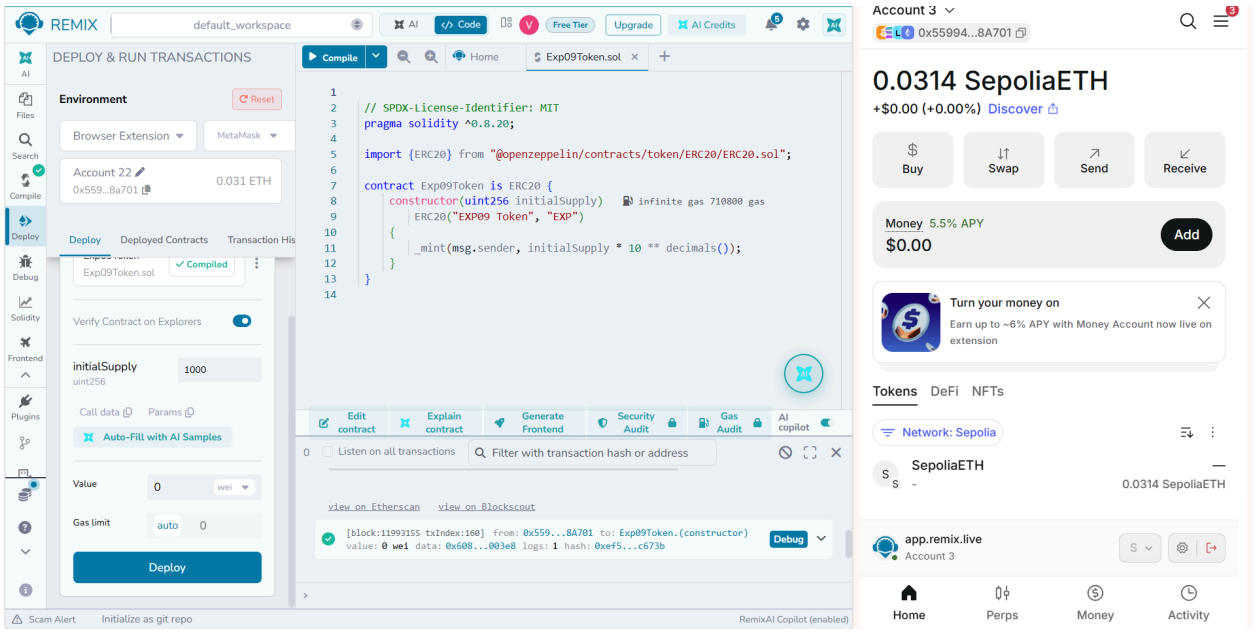
Solidity ERC-20 Smart Contract:



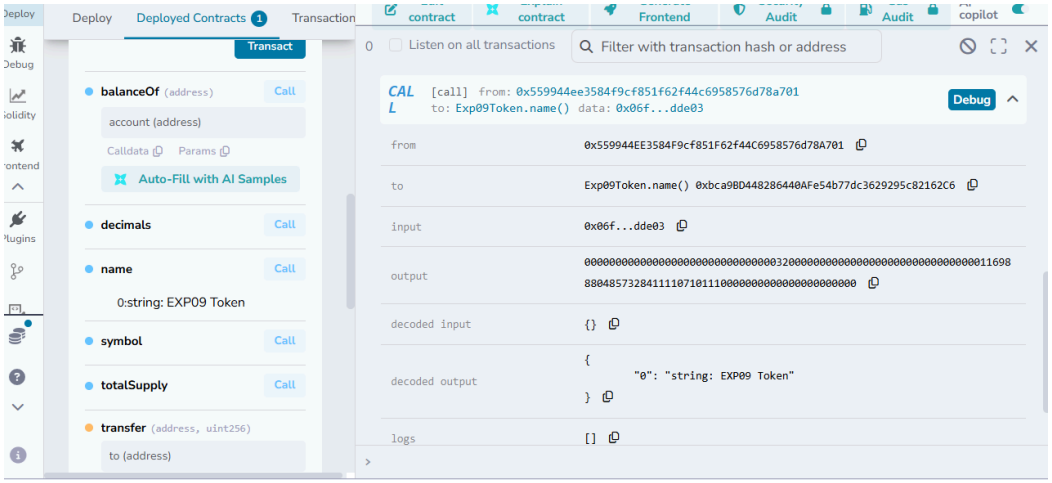
Contract Compilation:

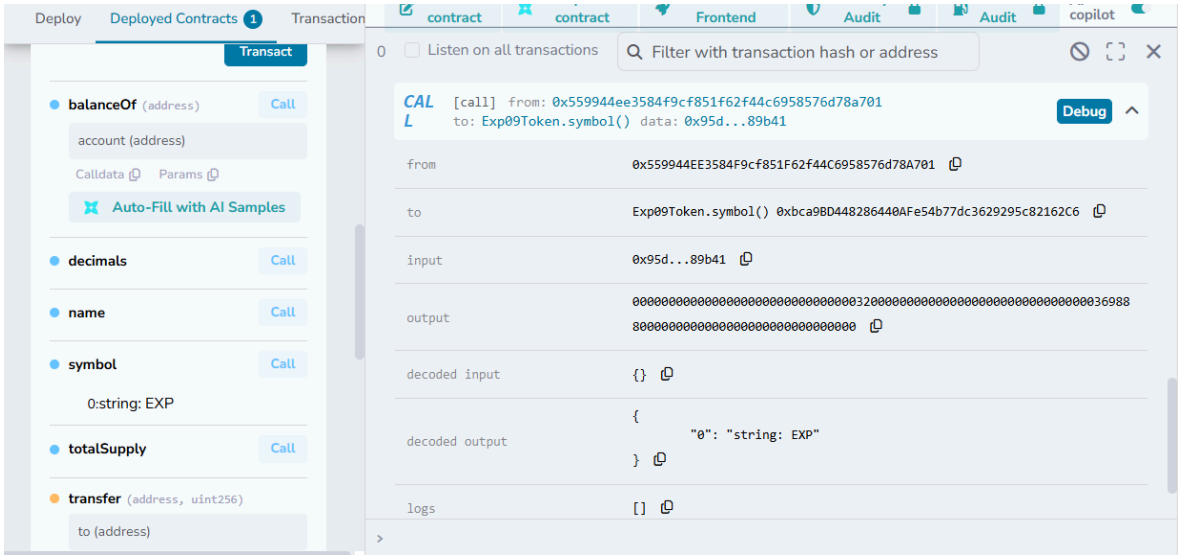


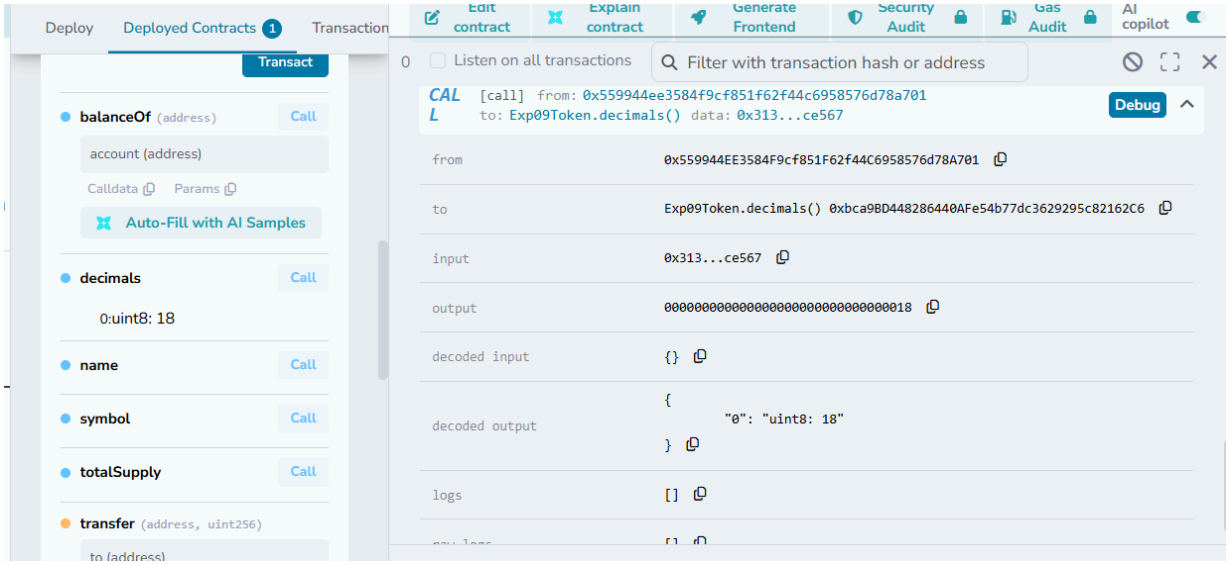
Contract deployment:

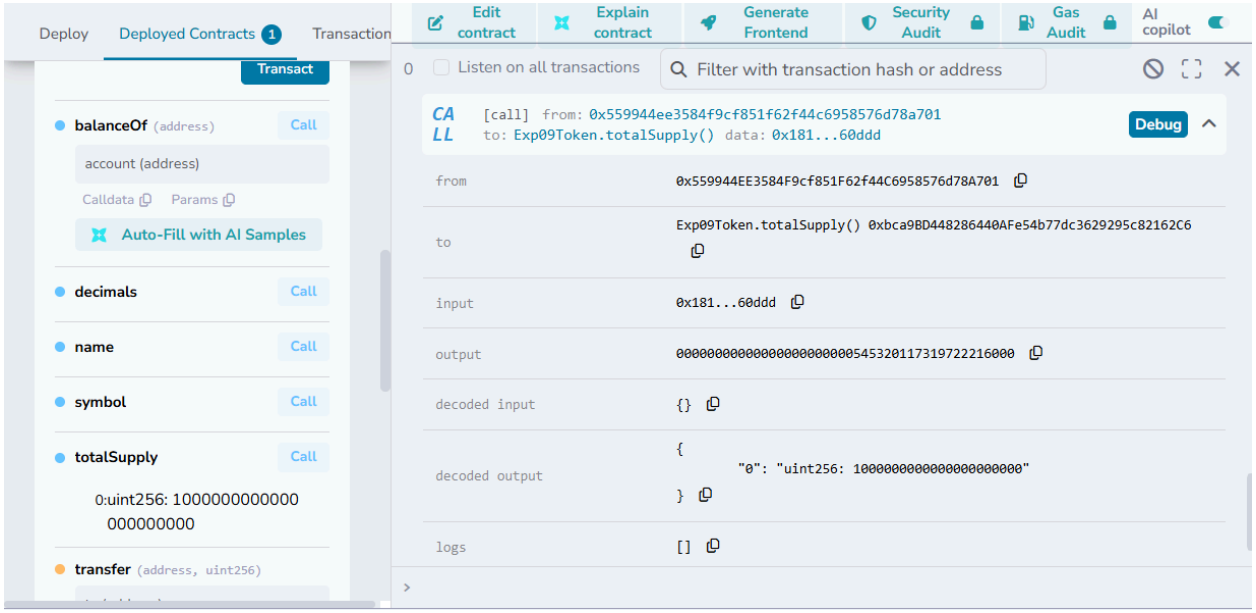


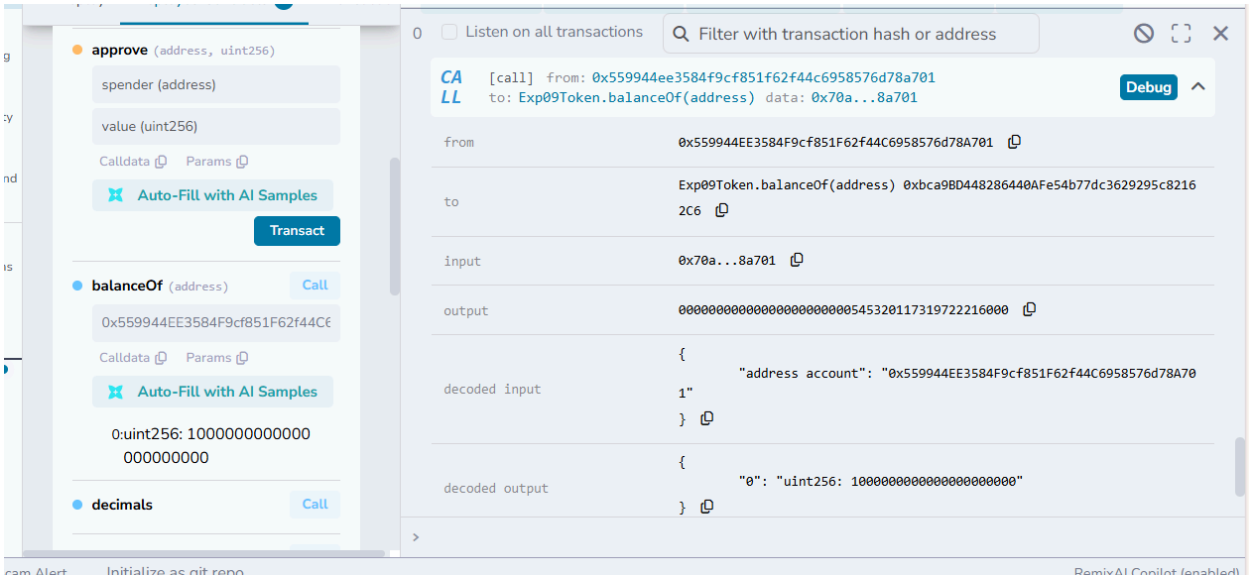
ERC-20 Function Execution:

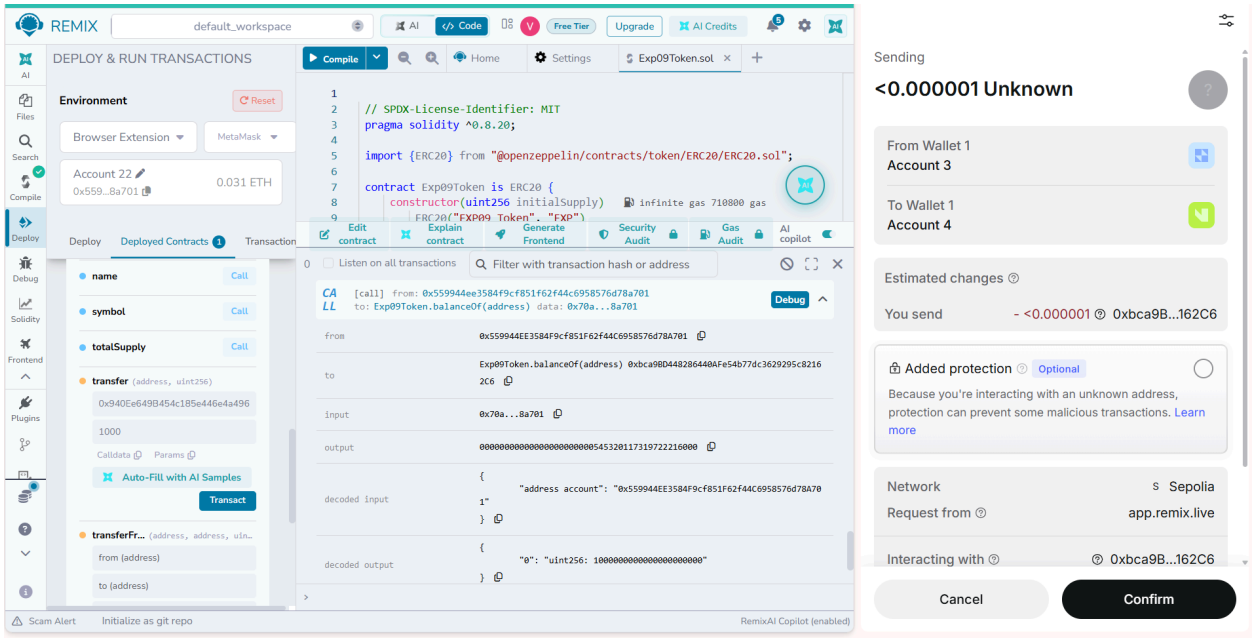

Deploy
Deployed Contracts 1
Transaction
Transact
balanceOf (address)
Call
account (address)
Calldata
Params
Auto-Fill with AI Samples
decimals
name
0:string: EXP09 Token
symbol
totalSupply
transfer (address, uint256)
to (address)
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL
[call] from: 0x559944ee3584f9cf851f62f44c6958576d78a701
to: Exp09Token.name() data: 0x06f...dde03
Debug
from
0x559944EE3584f9cf851F62f44C6958576d78A701
to
Exp09Token.name() 0xbca9BD448286440AFe54b77dc3629295c82162C6
input
0x06f...dde03
output
decoded input
{}
decoded output
{
"0": "string: EXP09 Token"
}
logs
[]

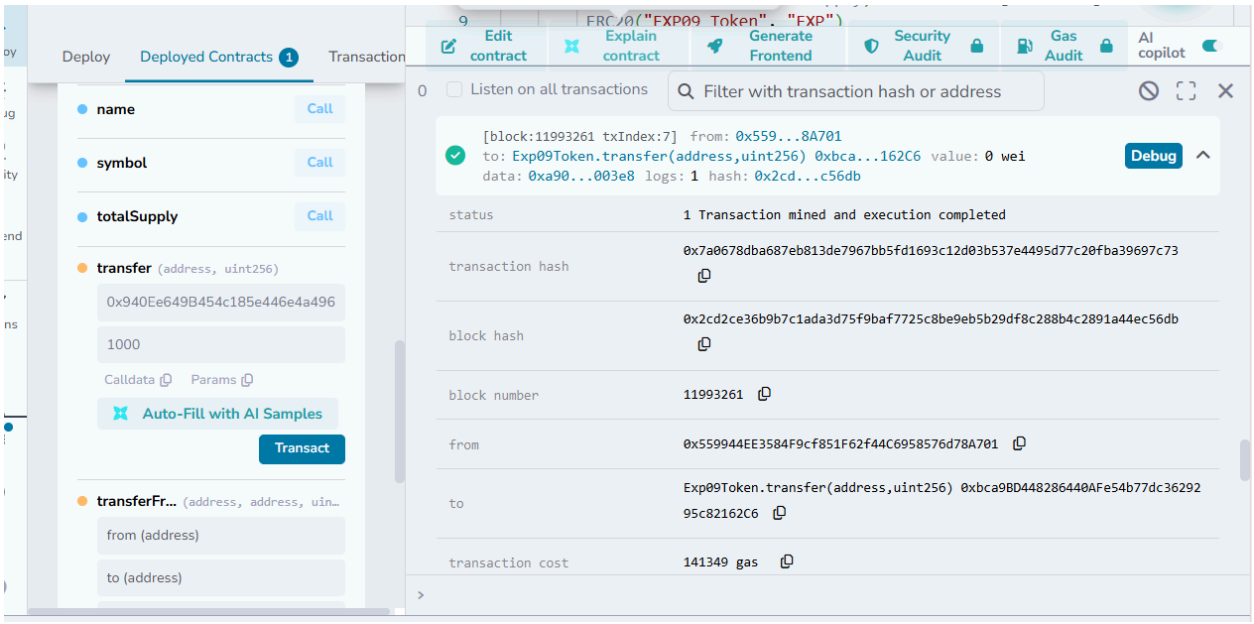

Deploy
Deployed Contracts 1
Transaction
Transact
balanceOf (address)
Call
account (address)
Calldata
Params
Auto-Fill with AI Samples
decimals
name
symbol
0:string: EXP
totalSupply
transfer (address, uint256)
to (address)
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL
[call] from: 0x559944ee3584f9cf851f62f44c6958576d78a701
to: Exp09Token.symbol() data: 0x95d...89b41
Debug
from
0x559944EE3584f9cf851F62f44C6958576d78A701
to
Exp09Token.symbol() 0xbca9BD448286440AFe54b77dc3629295c82162C6
input
0x95d...89b41
output
decoded input
{}
decoded output
{
"0": "string: EXP"
}
logs
[]

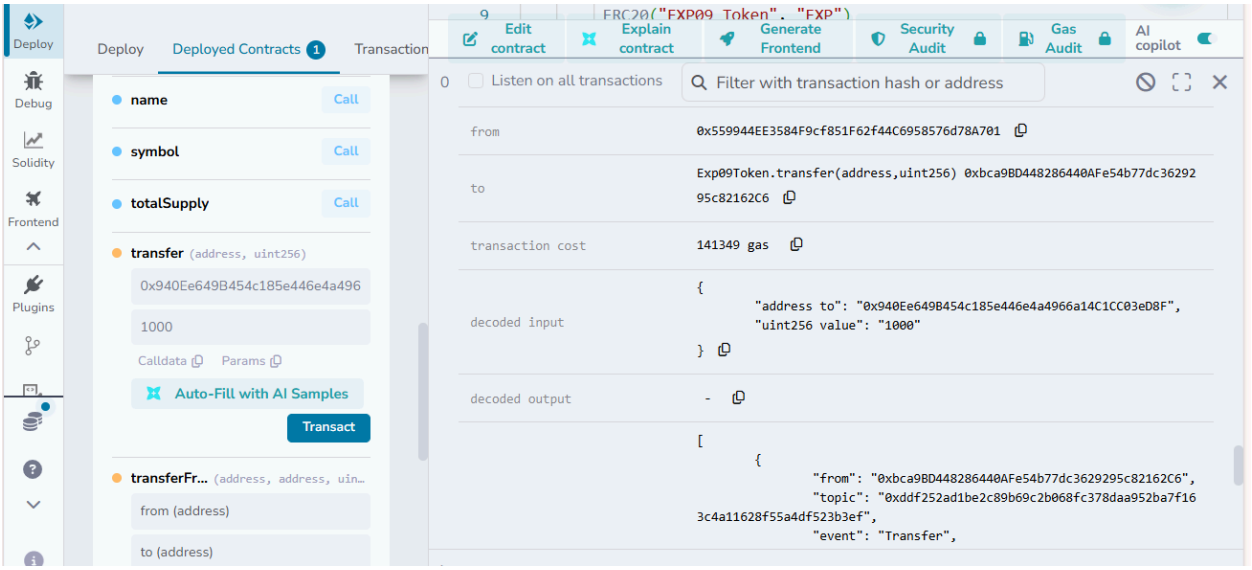

Deploy
Deployed Contracts 1
Transaction
Transact
balanceOf (address)
Call
account (address)
Calldata
Params
Auto-Fill with AI Samples
decimals
0:uint8: 18
name
symbol
totalSupply
transfer (address, uint256)
to (address)
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL
[call] from: 0x559944ee3584f9cf851f62f44c6958576d78a701
to: Exp09Token.decimals() data: 0x313...ce567
Debug
from
0x559944EE3584F9cf851F62f44C6958576d78A701
to
Exp09Token.decimals() 0xbca9BD448286440AFe54b77dc3629295c82162C6
input
0x313...ce567
output
decoded input
{}
decoded output
{
"0": "uint8: 18"
}
logs
[]

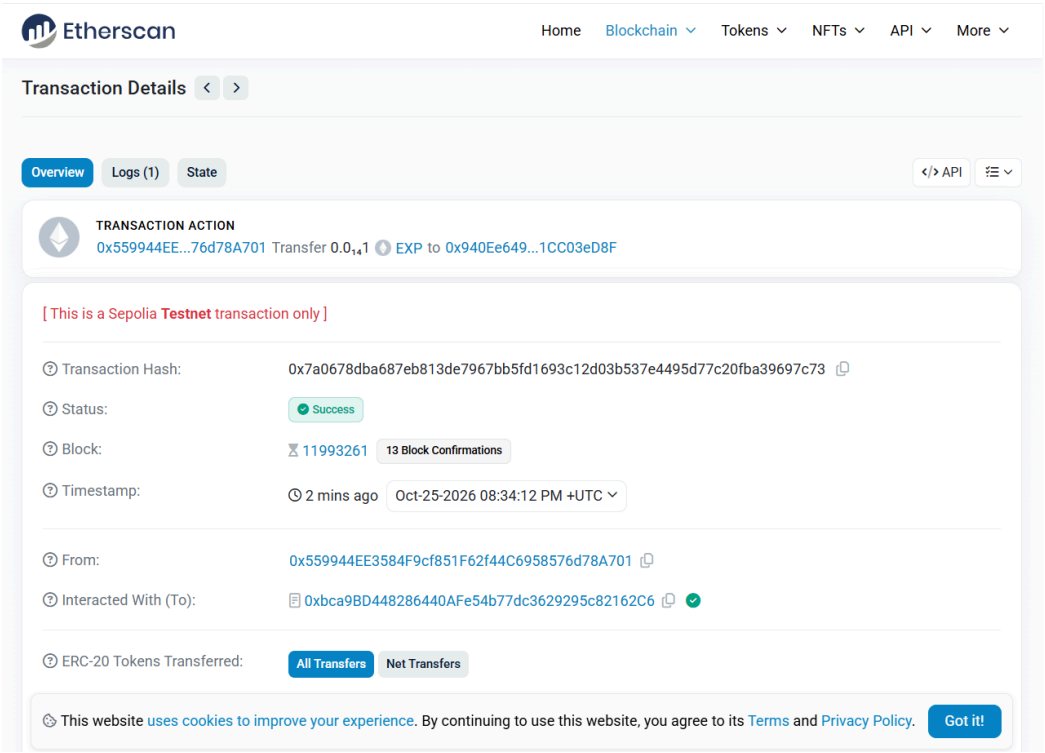

Deploy
Deployed Contracts 1
Transaction
Transact
balanceOf (address)
Call
account (address)
Calldata
Params
Auto-Fill with AI Samples
decimals
Call
name
Call
symbol
Call
totalSupply
Call
0:uint256: 1000000000000000000
transfer (address, uint256)
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
0
Listen on all transactions
Filter with transaction hash or address
CALL
[call] from: 0x559944ee3584f9cf851f62f44c6958576d78a701
to: Exp09Token.totalSupply() data: 0x181...60ddd
Debug
from
0x559944EE3584F9cf851F62f44C6958576d78A701
to
Exp09Token.totalSupply() 0xbca9BD448286440AFe54b77dc3629295c82162C6
input
0x181...60ddd
output
0000000000000000000000005453201173197222160000
decoded input
{}
decoded output
{
"0": "uint256: 1000000000000000000"
}
logs
[]


approve (address, uint256)
spender (address)
value (uint256)
Calldata
Params
Auto-Fill with AI Samples
Transact
balanceOf (address)
Call
0x559944EE3584F9cf851F62f44C6
Calldata
Params
Auto-Fill with AI Samples
0:uint256: 1000000000000000000
decimals
Call
0
Listen on all transactions
Filter with transaction hash or address
CALL
[call] from: 0x559944ee3584f9cf851f62f44c6958576d78a701
to: Exp09Token.balanceOf(address) data: 0x70a...8a701
Debug
from
0x559944EE3584F9cf851F62f44C6958576d78A701
to
Exp09Token.balanceOf(address) 0xbca9BD448286440AFe54b77dc3629295c82162C6
input
0x70a...8a701
output
0000000000000000000000005453201173197222160000
decoded input
{
"address account": "0x559944EE3584F9cf851F62f44C6958576d78A701"
}
decoded output
{
"0": "uint256: 1000000000000000000"
}


REMIX
default_workspace
DEPLOY & RUN TRANSACTIONS
Environment
Browser Extension
MetaMask
Account 22
0.031 ETH
Deploy
Deployed Contracts 1
Transaction
name
symbol
totalSupply
transfer (address, uint256)
Auto-Fill with AI Samples
Transact
transferFr... (address, address, uin...
from (address)
to (address)
Exp09Token.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
import {ERC20} from "@openzeppelin/contracts/token/ERC20/ERC20.sol";
contract Exp09Token is ERC20 {
constructor(uint256 initialSupply)
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
Debug
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
Sending
<0.000001 Unknown
From Wallet 1
Account 3
To Wallet 1
Account 4
Estimated changes
You send
- <0.000001
0xbca9B...162C6
Added protection
Optional
Because you're interacting with an unknown address, protection can prevent some malicious transactions. Learn more
Network
Sepolia
Request from
app.remix.live
Interacting with
0xbca9B...162C6
Cancel
Confirm


Deploy
Deployed Contracts 1
Transaction
name
symbol
totalSupply
transfer (address, uint256)
0x940Ee649B454c185e446e4a496
1000
Calldata
Params
Auto-Fill with AI Samples
Transact
transferFr... (address, address, uin...
from (address)
to (address)
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
[block:11993261 txIndex:7] from: 0x559...8A701
to: Exp09Token.transfer(address,uint256) 0xbca...162C6 value: 0 wei
data: 0xa90...003e8 logs: 1 hash: 0x2cd...c56db
Debug
status
1 Transaction mined and execution completed
transaction hash
0x7a0678dba687eb813de7967bb5fd1693c12d03b537e4495d77c20fba39697c73
block hash
0x2cd2ce36b9b7c1ada3d75f9baf7725c8be9eb5b29df8c288b4c2891a44ec56db
block number
11993261
from
0x559944EE3584F9cf851F62f44C6958576d78A701
to
Exp09Token.transfer(address,uint256) 0xbca9BD448286440AFe54b77dc3629295c82162C6
transaction cost
141349 gas





Conclusion:

The ERC-20 token was successfully created using Solidity and OpenZeppelin, compiled and deployed through Remix IDE on the Sepolia test network using MetaMask. The token's standard functions were tested to verify its name, symbol, decimal precision, total supply, account balances, and transfer functionality. Thus, the experiment demonstrated the creation, management, and transfer of a fungible blockchain token.